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Project Report

Computational Study of LTM in the Nocturnal Boundary Layer

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Abstract

Heat transfer processes that occur in the atmospheric boundary layer play an important role in many micrometeorological phenomena. The vertical temperature profile in the nocturnal boundary layer under clear and calm sky conditions is expected to be an inversion layer because of the lower thermal inertia of the ground in comparison to air. However, contrary to this, it was reported that the temperature minimum occurs a few decimeters above the ground. An aerosol-laden air layer close to the ground radiatively cools to the upper atmosphere faster than the ground itself, making the air layers close to the ground cooler. This phenomenon is called a 'Lifted Temperature Minimum (LTM)'. It results in the formation of a stable inversion layer above the LTM extending to several hundreds of meters. The formation of LTM and its impact on the atmosphere are being extensively studied. In this project, the LTM temperature profile was computationally simulated and an experimental (Unmanned Aerial Vehicle based) approach to study its effect in the atmospheric boundary layer was hypothesised.

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Part I

Computational Study of LTM in the Nocturnal Boundary Layer

Introduction

The heat transfer processes that take place in the nocturnal boundary layer play a very important role in determining the vertical temperature profile in the atmosphere. They also affect the occurrence of radiation fog and the formation of stable inversion layers. [\[1\]](#)[\[2\]](#)

The ground is a better emitter than air, suggesting that the ground should cool faster than the layers of air above it, causing a stable inversion layer to occur in the atmosphere. However, extensive experimental and theoretical research has shown that the temperature profile occurring in the atmosphere has its temperature minimum at a height of a few decimeters above the ground. This phenomenon was referred to as a 'Lifted Temperature Minimum' (LTM) and was originally discovered by Ramdas and is also referred to as the Ramdas paradox.[\[3\]](#) The conditions and factors upon which the vertical temperature variation follows an LTM profile have been studied theoretically as well as experimentally. It was determined that under homogenous atmospheric conditions (without the presence of aerosols), preferential cooling does not occur close to the ground, thus leading to a temperature profile which does not have an LTM.[\[3\]](#) Studies were also performed to analyse the dependency of the LTM formation on humidity [\[1\]](#), and wind speed.[\[4\]](#)[\[5\]](#)

The temperature profile featuring the LTM can be described by the height at which the

LTM occurs above the ground (the LTM height) and the difference between the temperature of the ground and the temperature minimum (the LTM intensity).

1.1 Objectives

- To computationally simulate the LTM profile occurring in the NBL under calm and clear sky conditions and in the absence of advection. In an idealised set-up assuming uniform size distribution of aerosols.
- To study the variation of LTM characteristics due to the change in the concentration of aerosols.
- To estimate the time constant for the transient response of a system of aerosols radiatively cooling to the upper atmosphere leading to the formation of an LTM profile.
- To investigate the radiative and conductive flux divergences occurring in the system.

1.2 Factors Affecting LTM Formation

The primary reason for the formation of an LTM profile in the atmosphere was determined to be the heterogeneity caused by the presence of aerosols, thus the characteristics of the aerosol present in the atmosphere such as its concentration, emissivity and size distribution play a crucial role in determining the nature of the LTM.

The dependence on the humidity of the atmosphere was found to be absent by performing field experiments.[1] The dependency on wind speed is more prominent, as the LTM intensity diminishes with the increase in wind speed. It was observed that for

high enough wind speeds, the LTM profile disappeared altogether. Most cases of LTM formation has been observed when the wind speed is less than 2 m/s.[\[5\]](#)

1.3 Energy Equation

The heat transfer equation for the energy exchange occurring in the atmosphere can be written as follows

$$\rho C_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial^2 y} - q''' \quad (1.1)$$

Here, ρ and C_p represent the density and specific heat capacity of air respectively. T is the air temperature, y is the coordinate used to represent the vertical distance above the ground and k is the thermal conductivity of air. q''' is the flux divergence term that captures the radiative cooling of the system to the upper atmosphere. The radiative flux divergence depends on the size of the aerosol particles, their number density and emissivity. q''' can be written in terms of these factors as mentioned in [section 2.1](#).

1.4 Time Response due to Aerosols

The transient response of a initially homogenous system ([figure 3.1](#)) to the presence of aerosols can be characterised by a time constant τ . With the inclusion of an aerosol layer, temperatures start to fall and the vertical temperature profile moves towards one with an LTM. The time scale for this process can be expressed as the time constant for an exponential curve, which would be the time taken to achieve 63.2% of the final LTM intensity.

The transient temperature change can be expressed as an exponential function dependent

on τ by using an integral heat balance and assuming the instantaneous conduction flux to vary with the change in temperature linearly.[\[1\]](#)

$$\Delta T = \Delta T_o(1 - e^{-\frac{t}{\tau}}) \quad (1.2)$$

Here, ΔT and ΔT_o represent the change in temperature at time t and total change in temperature respectively.

Governing Equation

The general form of the heat transfer equation consists of convection, conduction and any other heat source terms as given below

$$\rho C_p \frac{\partial T}{\partial t} - \nabla(k \nabla T) + (\bar{u} \cdot \bar{\nabla})T = q''' \quad (2.1)$$

However, in the simulation implemented, it is assumed that only conduction and radiation are present. Moreover, the problem at hand was two-dimensional (figure 3.1), with significant temperature variation only in the vertical (y-coordinate) direction. Thus the governing equation that was numerically solved is equation 1.1

2.1 Radiative Cooling

The q''' term in equation 1.1 captures the radiative heat transfer in the system due to the presence of aerosols. It can be expressed as

$$q''' = N \times Q_R \quad (2.2)$$

Here, the radiative flux divergence q''' is expressed in terms of the aerosol number density, N and the radiative cooling of a single aerosol particle, Q_R . The radiative cooling of a single aerosol particle to the upper atmosphere can be expressed in terms of its surface

area and emissivity [6][1]

$$Q_R = -\frac{\sigma A}{2} \epsilon_p (\epsilon_g T_g^4 + T_{sky}^4 (2 - \epsilon_g) - 2T_p^4) \quad (2.3)$$

where, σ denotes the Stefan Boltzmann constant, A is the surface area of the particle, T_g , T_p and T_{sky} denotes the temperature of the ground, aerosol particle and upper atmosphere respectively and ϵ_g and ϵ_p represent the emissivities of the ground and aerosol particle respectively. Aerosol particles of uniform diameter of 1 μm were considered for this simulation. The temperature of the ground is taken to be 300K and that of the upper atmosphere is taken to be 285K. The upper atmosphere is observed to be at least 10-15 K cooler than the ground. ϵ_p and ϵ_g are taken to be 0.9 each. In order to obtain the net radiative cooling, Q_R needs to be multiplied by the number density of aerosols present in the system. Moreover, particle-particle interactions between aerosols were neglected while expressing the radiative forcing term.

2.2 Number Density of Aerosols

The number density of aerosols present in the atmosphere directly affects the magnitude of radiative cooling that occurs. A non-uniform vertical distribution of aerosol number density was found to be the primary reason for the observed hyper-cooling near the ground. The particle density variation is taken to be [6]

$$N = \frac{N_p}{A} \times [1.4 \exp\left(\frac{-y}{0.05}\right) + 0.03] \times 10^{12} \text{ m}^{-3} \quad (2.4)$$

Here, N_p is a parameter that has been varied in the simulation to change the number of aerosols present. The order of magnitude of the aerosol number density varies from a minimum of 10^8 to a maximum of 10^{13} in the system considered in this study. This

has been further mentioned in section 3.2. Figure 2.1 shows the variation of N with the vertical distance.

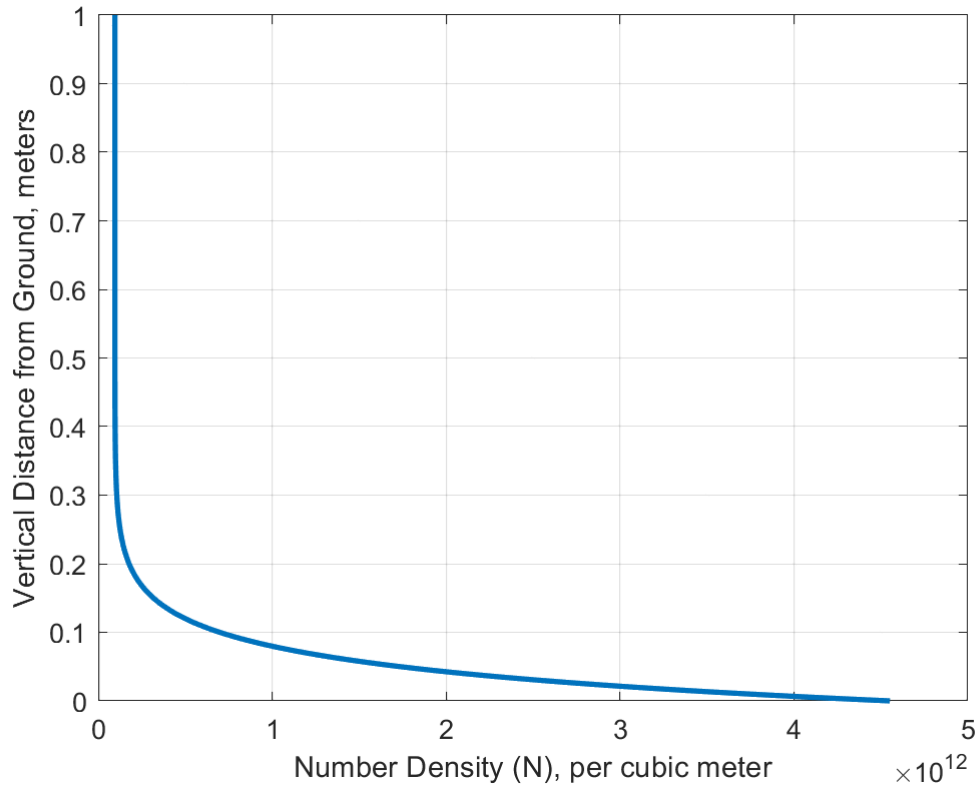


FIGURE 2.1: Aerosol Number Density Variation

Computational Set-Up

The simulation was modelled and run on MATLAB using its partial differential equation toolbox. The simulation solved the PDE given in equation 1.1 after incorporating the radiative flux divergence term as mentioned in section 2.1. The solver's algorithm uses the Finite Element method to solve the partial differential equation. [7]

3.1 Geometry

The domain to conduct the study was kept to be a rectangular region with a height of 1 m and a width of 0.75 m. Dirichlet boundary conditions were applied to the temperature at the top and bottom edges. The temperature at the top edge was fixed at 302K and the bottom edge's temperature was fixed as the ground temperature T_g . On the other two edges, a zero flux condition was imposed. The meshing of the domain was carried out with triangular meshes, with a maximum height of 0.01 m in the entire domain and 0.0001 m near the ground (bottom edge of the geometry).

The set-up was made to simulate the LTM formation that was achieved experimentally [1], thus the temperature was fixed in the top and bottom edges of the domain, representing the top and bottom constant temperature boundaries of the experimental set-up. The T_{sky} term in the governing equation accounts for the radiative heat transfer of the system with the upper atmosphere.

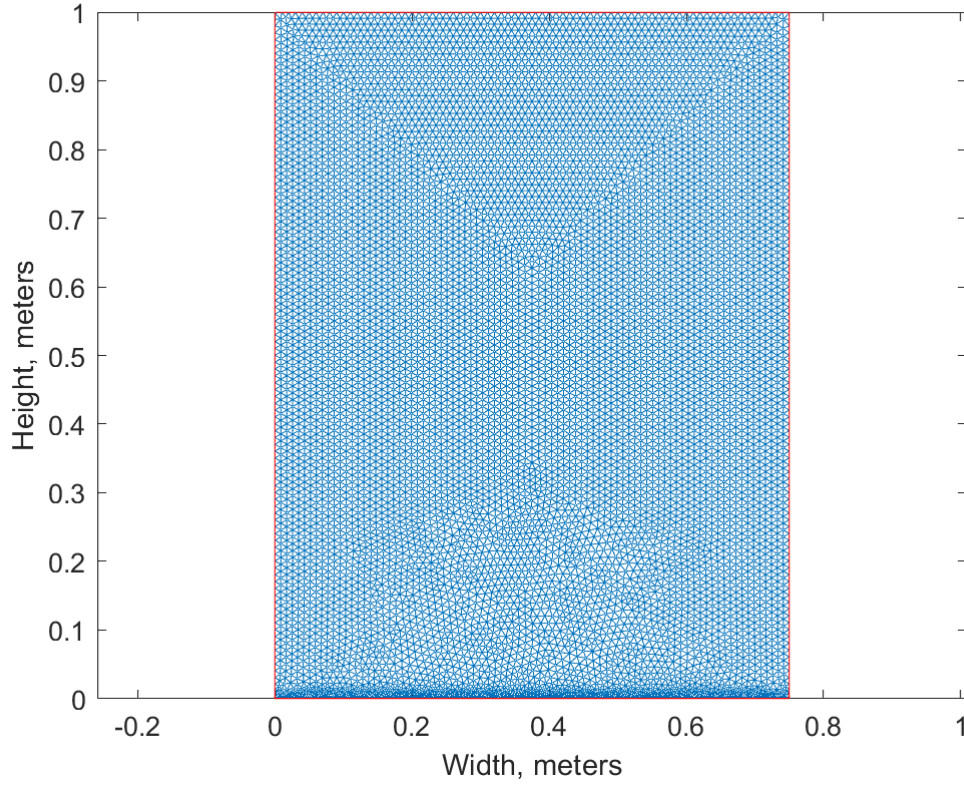


FIGURE 3.1: Domain Geometry and Meshing

3.2 Variation of Number Density

The number density of the aerosols in the domain has been varied according to the equation 2.4. The values of parameter N_p is given in table 3.1. Here N_b and N_t are number densities at the top and bottom of the rectangular domain.

N_p	$N_b \text{ (m}^{-3}\text{)}$	$N_t \text{ (m}^{-3}\text{)}$
0.1	4.55×10^{10}	9.54×10^8
1	4.55×10^{11}	9.54×10^9
10	4.55×10^{12}	9.54×10^{10}
25	1.14×10^{13}	2.38×10^{11}

TABLE 3.1: Aerosol Number Density Variation

Results and Analysis

4.1 Stationary Solution

The stationary solution for the temperature profile of the system was first obtained. The variation of temperature over the surface of the domain and the variation with the vertical distance from the ground for $N_p = 0.1$ are given in figures 4.1 and 4.2

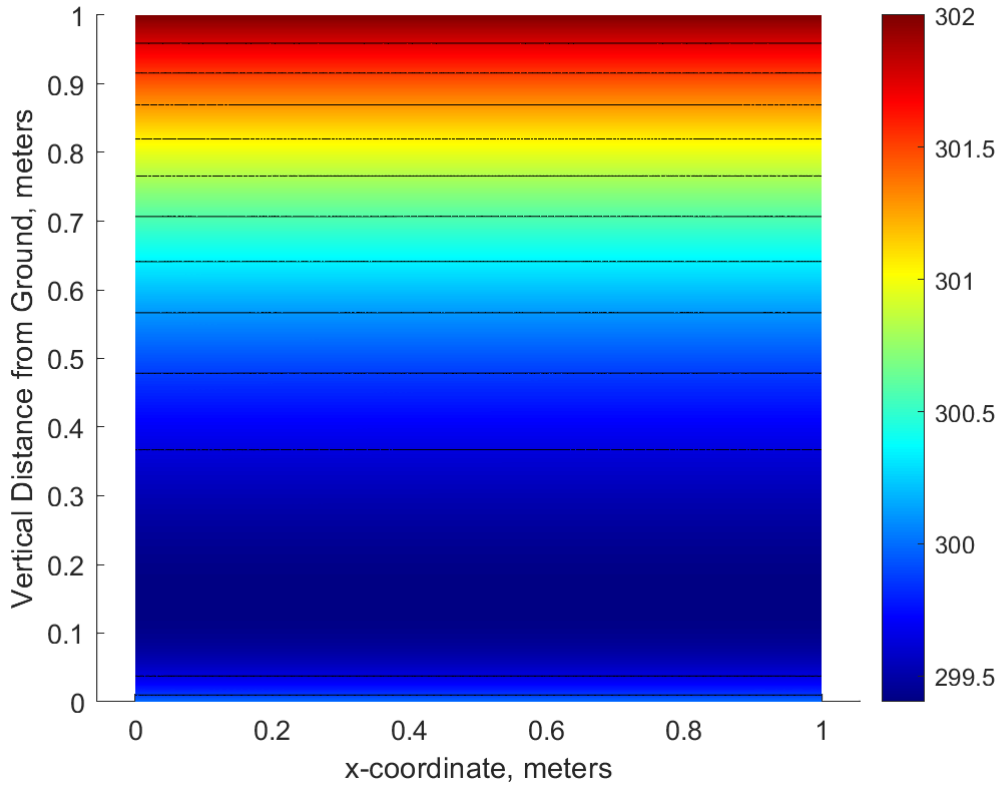
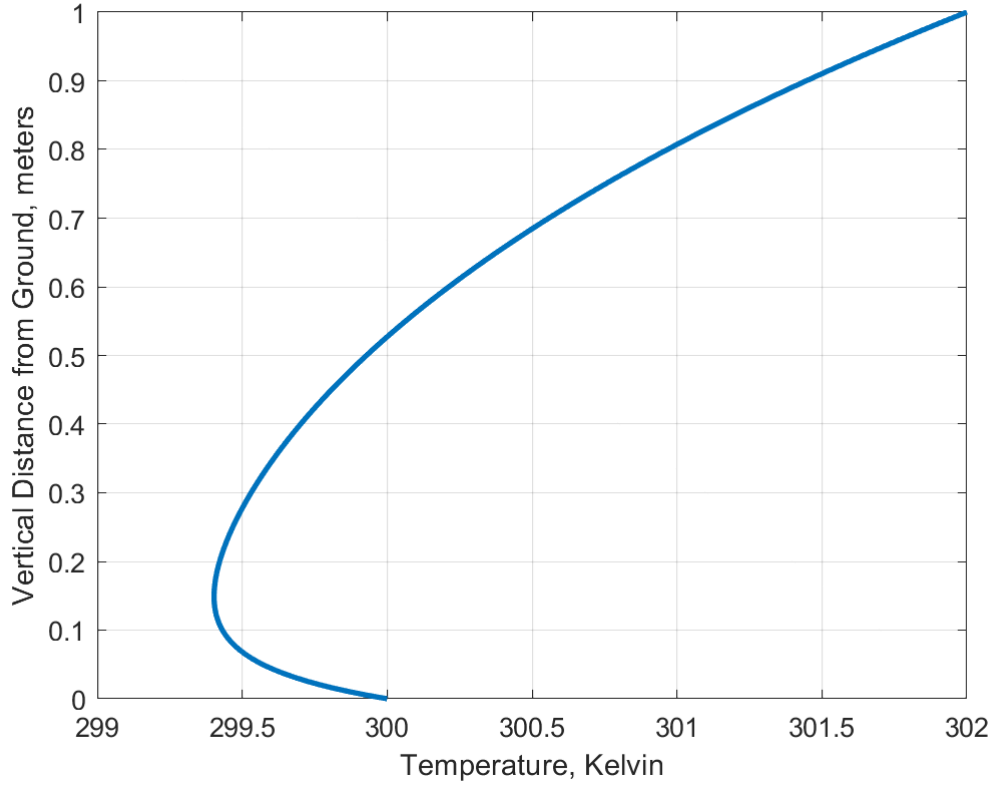


FIGURE 4.1: Surface Temperature Variation for $N_p = 0.1$

FIGURE 4.2: Vertical Temperature Profile for $N_p = 0.1$

From the figures above it is apparant that with an increase in height from the ground, the temperature drops to reach a minumum and then increases forming an inversion layer. From figure 4.1 it is evident that there is no variation of the temperature in the x-direction, which is expected as there is no advection in the system.

The variation of LTM intensity for different values of N_p was also studied, the results are shown in table 4.1 and the temperature profiles are shown in figure 4.3.

N_p	LTM Intensity (K)	LTM Height (cm)
0.1	0.60	14.5
1	3.77	18.5
10	7.918	29

TABLE 4.1: Variation of LTM Characteristics with Number Density

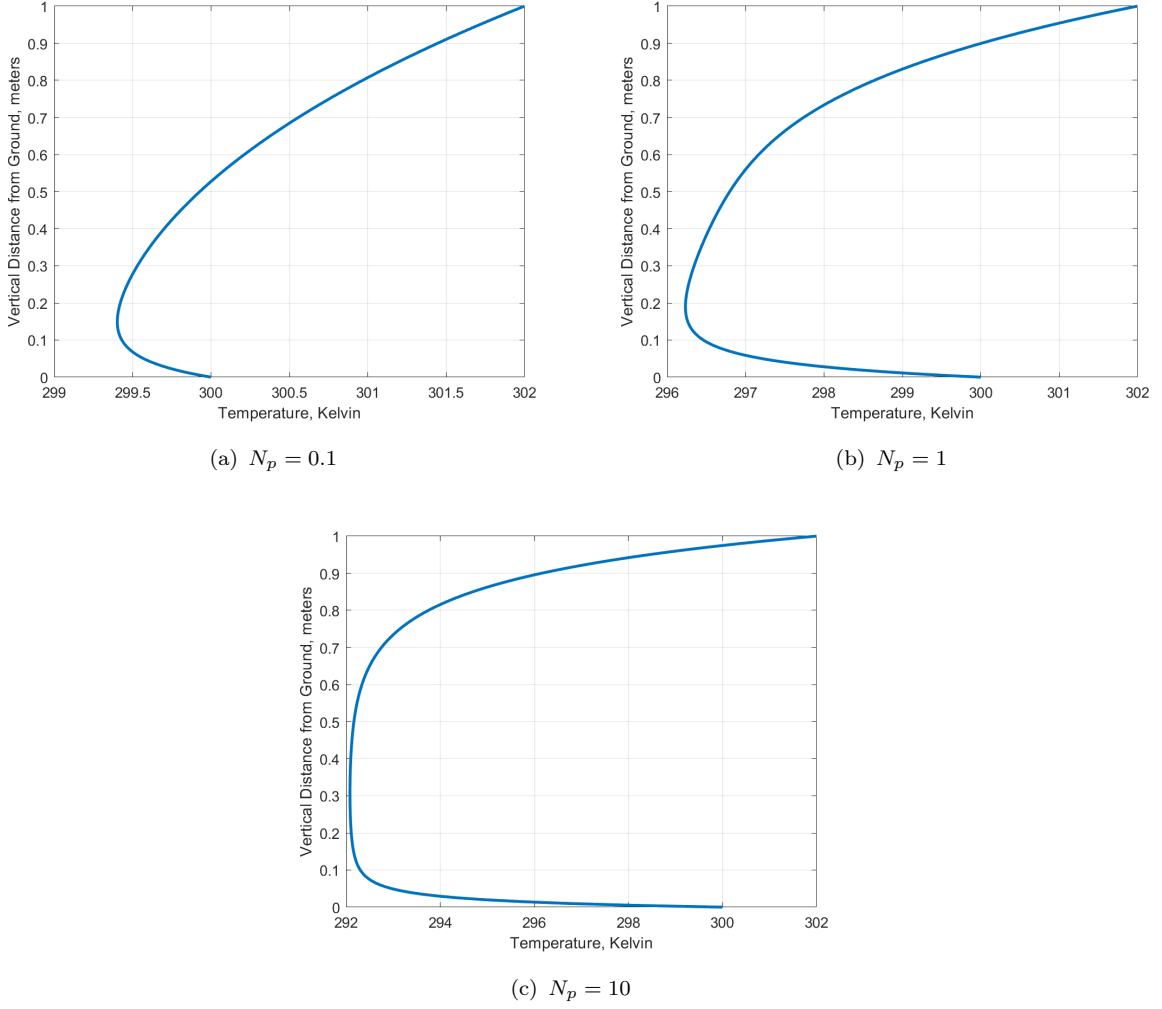


FIGURE 4.3: Variation of Temperature Profile with Number Density

It was observed that both the LTM intensity and the LTM height increase with the increase in the value of the parameter N_p . With an increase in the number density of the aerosol, the hyper-cooling near the surface increases causing the temperature to drop even more, thus increasing the LTM intensity. With an increase in aerosol concentration, the balance of conductive and radiative flux occurs in such a way that the LTM is formed farther away from the ground.

4.2 Transient Solution

The system was initially assumed to be in a steady state in an homogenous atmosphere. In the absence of convection and radiation, the initial condition of the system followed a

linear vertical temperature profile. The radiative forcing term was introduced at $t = 0$ and the simulation was run until the system reached the steady state LTM profile discussed in section 4.1. The initial and steady state temperature profile for $N_p = 0.1$ is shown in figure 4.4.

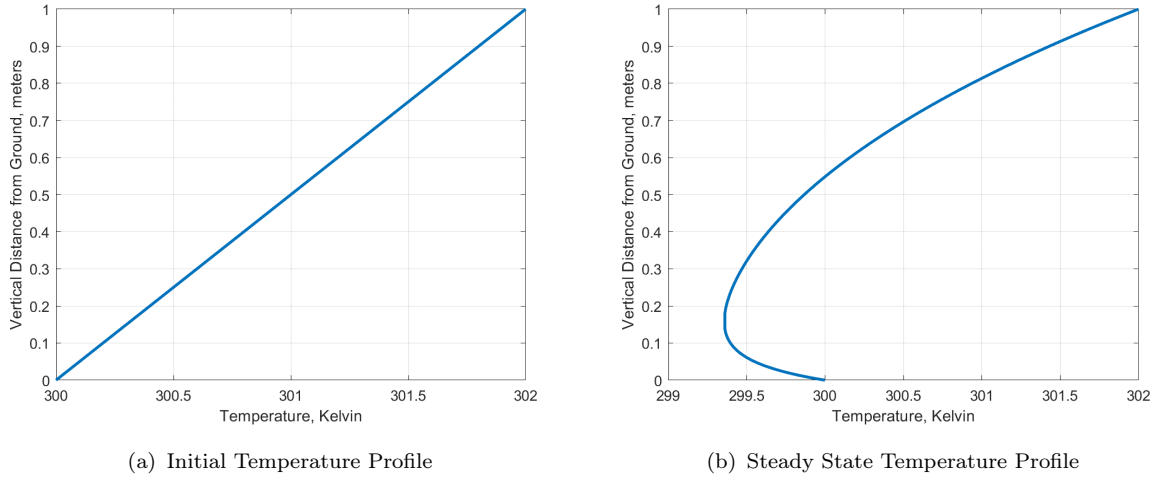


FIGURE 4.4: Temperature Profiles obtained from Transient Modelling

The simulation was run for 30,000 seconds, and the steady state temperature profile achieved was verified to be identical to the stationary solution as shown in figure 4.5.

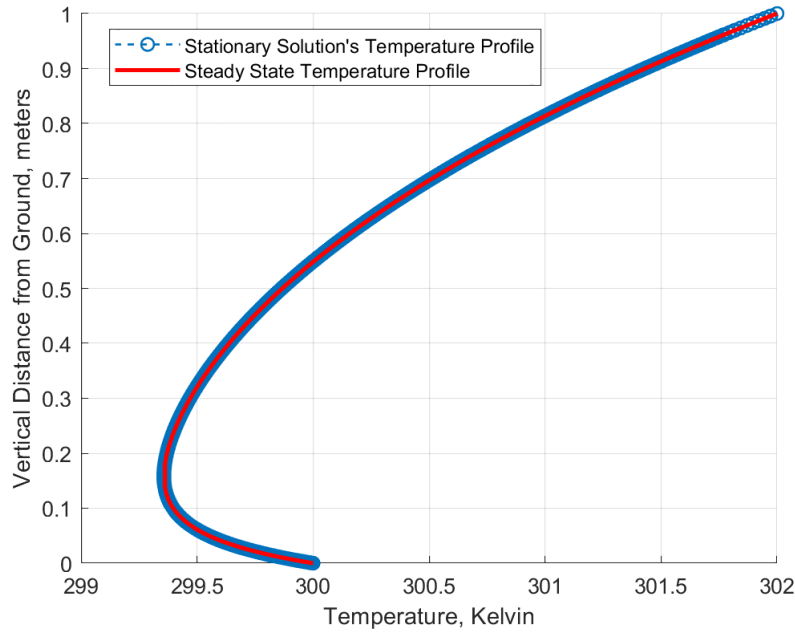
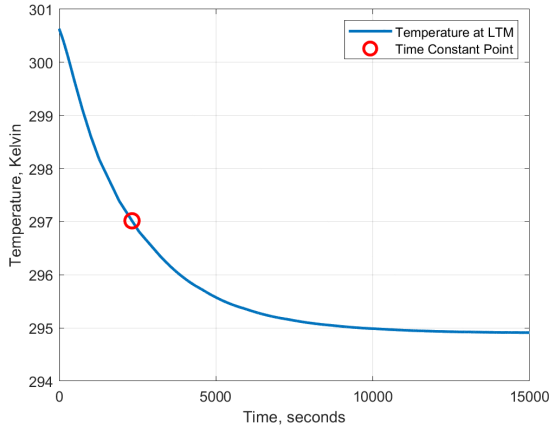


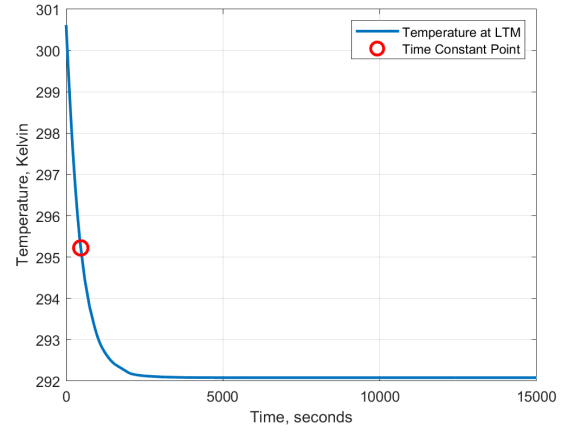
FIGURE 4.5: Comparison of stationary and steady state transient solution's temperature profile for $N_p = 0.1$

4.3 Time Constant

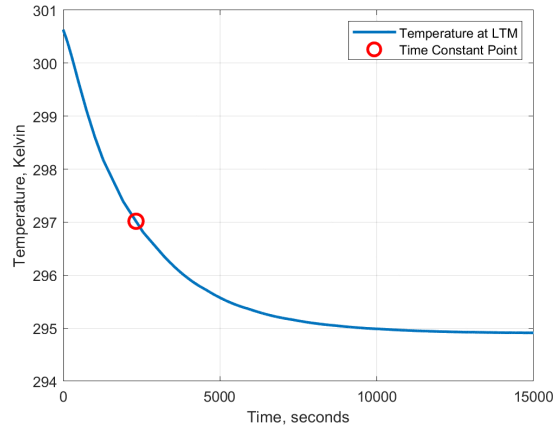
As mentioned in section 1.4, the time constant of the transient system can be captured as the time taken to achieve 63.2% of the temperature change. The time constant of the system for different values of N_p is given in table 4.2 and the response curve at the LTM for each case is shown in figure 4.6.



(a) $N_p = 1$



(b) $N_p = 10$



(c) $N_p = 25$

FIGURE 4.6: Transient Response of the System for Different Values N_p

N_p	Time Constant (s)
1	2364
10	467
25	190

TABLE 4.2: Variation of Response Time of the System for Different Values of N_p

It can be seen from the graphs in figure 4.6 that the time taken to achieve the steady state LTM profile decreases with the increase in the concentration of aerosols. This can be inferred from the fact that the larger the number of aerosols present, the more the radiative cooling near the ground, leading to a faster drop in the temperatures to an LTM profile.

4.4 Flux Divergence

The radiative and conductive flux divergences were found according to the equation 1.1. Where q''' is defined in equation 2.2.

The variation of the radiative and conductive flux divergence were plotted against the vertical distance from the ground, to analyse the nature of the heat transfer processes that occur in the atmosphere. The net divergence flux was also calculated from both the divergence fluxes. The net flux divergence drives the hyper-cooling close to the ground in the nocturnal boundary layer. The radiative flux divergence signifies the cooling due to the transfer of energy from the aerosol layers to the upper atmosphere, playing a crucial role in the formation of an LTM.

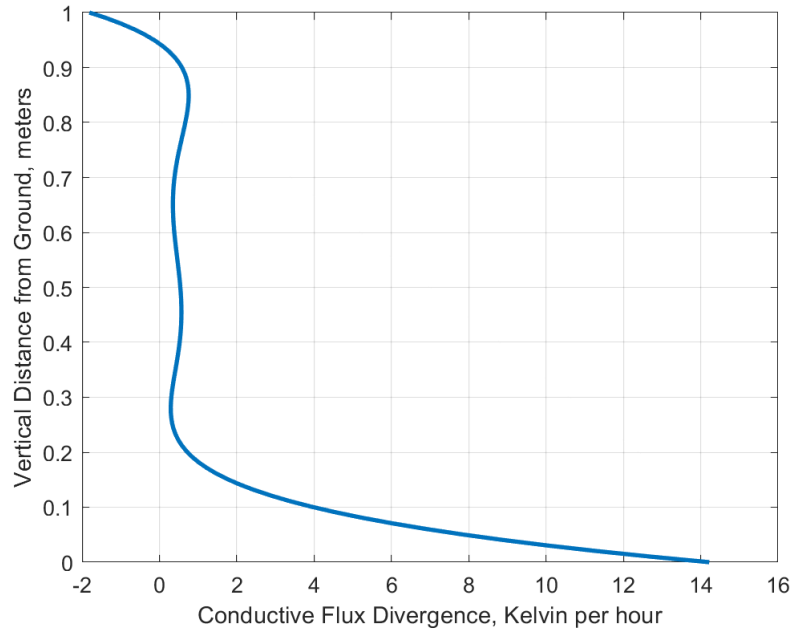


FIGURE 4.7: Variation of Conductive Flux Divergence with Vertical Distance for $N_p = 0.1$

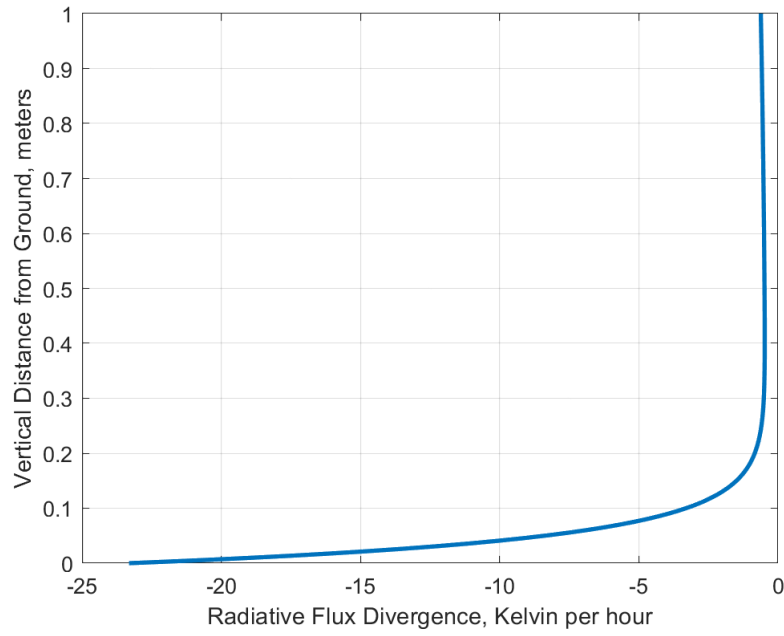


FIGURE 4.8: Variation of Radiative Flux Divergence with Vertical Distance for $N_p = 0.1$

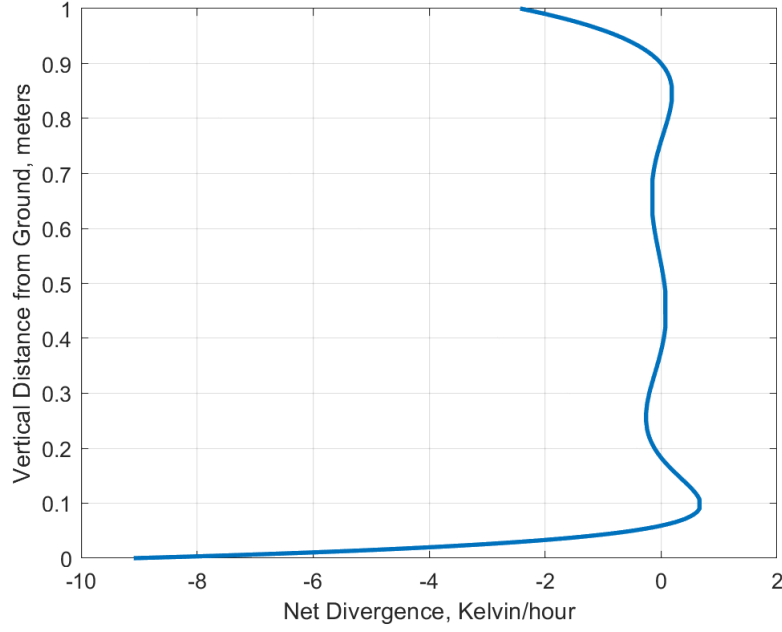


FIGURE 4.9: Variation of Net Flux Divergence with Vertical Distance for $N_p = 0.1$

The nature of the plot of the flux divergence gives an insight on the net cooling that is occurring in the system. The divergence is seen to decrease in magnitude as the vertical distance from the ground increases, suggesting a strong near-ground cooling. The flux divergence is close to zero at larger vertical distances, indicating very little cooling. This can be observed in the approximately linear inversion layer in the temperature profile above the LTM.

Conclusions

An computational study of LTM in the atmospheric boundary was conducted in a two-dimensional domain. The results of the simulation showed the presence of a lifted temperature minimum occurring a few decimeters above the ground, in the presence of aerosols. The LTM intensity was observed to increase with an increase in the concentration of aerosols, this was due to the added cooling by the additional aerosol particles. It varied from 0.5 - 8 K and the LTM height was seen to vary from 14 to 30 cm, for the aerosol number densities close to the ground having orders of magnitudes between 10^{10} and 10^{13} .

In the absence of radiative heat transfer in the system, the temperature profile obtained was linear owing to the heat transfer due to conduction. When the aerosols were introduced into such a system, there was radiative cooling with the upper atmosphere. The system thus followed a transient response to achieve a temperature profile with an LTM in the steady state. The curve of the transient response obtained was approximately exponential, and could be characterised by its response time. This response time for different aerosol concentrations was studied. Moreover, it was observed that the time taken to reach the steady state decreases for the increase in the number density of aerosols, due to the increased radiative interaction to the upper atmosphere leading to faster cooling near the ground.

The heat transfer processes occurring in the system were limited to conduction and radiation. Each of their flux divergences was determined. The radiative flux divergence

was seen to decrease in magnitude drastically as the height from the ground increased. The curve followed a similar characteristic to the curve denoting the variation of the number density of aerosols with height. The simulation aimed at computationally recreating the laboratory set-up aimed at studying LTM profiles and the boundary conditions for the model were chosen accordingly. The results obtained for the vertical temperature variation reiterated the experimental results which indicate the LTM formation and its characteristics.[\[1\]](#)

Part II

Calibration and Testing of Thermistor-Based Temperature Sensors

Introduction

The LTM is seen to occur under calm and clear sky conditions when an aerosol layer is present over the ground surface. The phenomenon leads to the formation of a stable inversion layer which extends up to several hundreds of meters. Understanding the effect of this phenomenon on the vertical temperature profile requires an accurate measurement of the temperature at every altitude in the atmosphere for at least a few hundred meters. Various methods have been used to map this temperature profile. However, many of them are immobile and cannot be used for measurements at higher altitudes. Other methods like weather balloons with radiosondes, can collect data from the atmosphere even at higher altitudes but are not very accurate due to their inability to hover and perform a controlled flight.

An unmanned aerial vehicle or UAV is an aircraft that is capable of performing a partially or fully autonomous flight. A UAV can be further classified as a fixed-wing aircraft and rotorcraft. Rotorcrafts such as quadcopters and hexacopters possess the ability to perform a purely vertical flight including vertical take-offs and landings. They are also capable of hovering at a certain altitude. A rotorcraft was thus considered one of the most accurate methods of measurement for this purpose. Its ability to perform a vertically controlled flight and hover at desired altitudes makes it an ideal candidate for conducting experimental research to find accurate atmospheric data for higher altitudes. The experiment devised was to use a UAV with all required sensors (for temperature,

relative humidity and wind velocity) to get accurate atmospheric data corresponding to a wide range of altitudes.

Different atmospheric sensors were decided to be used along with the UAV to capture the required data. The most important of them is the sensors to measure temperature. Other sensors were for humidity and wind velocity measurements. These sensors were to be mounted at a height of a few centimetres above the body of the drone to avoid errors in the data from the heating of drone circuitry and the motion of its propellers. Anemometers were researched for the measurement of wind velocity and thermistors, thermocouples, digital temperature sensors, etc for temperature measurement.

Thermistors are resistance-based temperature measurement devices. The resistance of a thermistor changes with temperature and the change in resistance can be determined by measuring the characteristics of an appropriate circuit. In this project, Negative Temperature Coefficient or NTC thermistors were used. The resistance of these thermistors decreases with an increase in temperature. Thermistors are cheap, easily available, relatively accurate and fast at sensing temperature change, making them a possible choice for a UAV-based atmospheric study.

One of the biggest challenges in designing the above-mentioned experiment was the requirement for a very fast response sensor. The UAV can be expected to hover at a particular altitude for a very limited amount of time, making it necessary for the sensors to have a very fast response time. The response time of a thermistor can be characterised based on its time constant. This quantity was determined for various thermistors to check their utility for the experiment.

Set-Up and Methodology

7.1 Apparatus

Three NTC thermistors A, B and C were used for the testing purposes. The specifications for each are mentioned below in table 7.1.

Thermistor	Resistance at 25°C (k Ω)
A	10
B	2.2
C	10

TABLE 7.1: Specifications of Thermistors

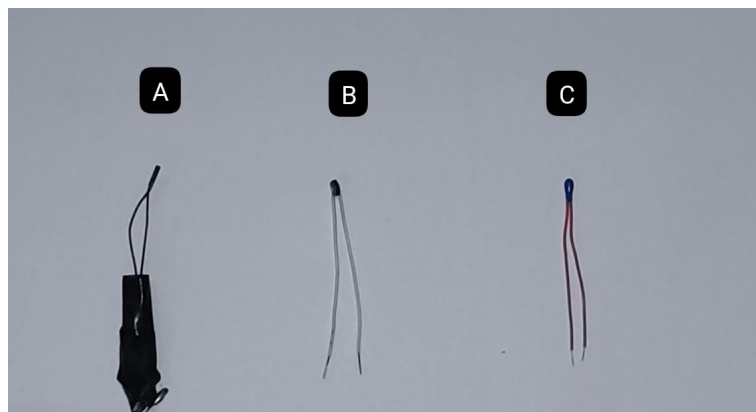


FIGURE 7.1: Thermistors Used

The thermistors were connected to a Keysight 34972A LXI data logger. The data of the variation of the resistance of the thermistor with time was determined. The temperature of the system was measured using a digital thermometer.

7.2 Calibration

The resistance of each of the three thermistors was noted at room temperature. A hot water bath and an ice bath of known temperatures were prepared, the thermistors were dipped into these, and the change in their resistance was noted.

The Steinhart-Hart equation as given below was used to determine the relation between the temperature and the resistance of each of the thermistors.

$$\frac{1}{T} = A + B(\ln R) + C(\ln R)^3 \quad (7.1)$$

With the resistance of each thermistor at each of the three known temperatures (the ambient, the ice bath, the hot water bath) available, the coefficients of the Steinhart-Hart relation were determined by solving the following matrix equation

$$\begin{bmatrix} 1 & \ln R_1 & \ln^3 R_1 \\ 1 & \ln R_2 & \ln^3 R_2 \\ 1 & \ln R_3 & \ln^3 R_3 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} \quad (7.2)$$

Here, R_1 , R_2 , R_3 represent the resistance of a thermistor at temperatures T_1 , T_2 and T_3 respectively. The calibration charts for each of the three thermistors were plotted with the observed temperature-resistance relationship along with the ideal relationship.

The coefficients obtained from calibrating these sensors were subsequently used to determine their time constants.

7.3 Time Constant

In order to obtain accurate data from a UAV, it is necessary that the sensors have a very fast response time due to the limited period for which the drone can hover. The response time of a sensor can be characterized by its time constant, τ . Thermistors follow an exponential response curve given by the following equation

$$T - T_1 = (T_2 - T_1)(1 - \exp\left(\frac{-t}{\tau}\right)) \quad (7.3)$$

Here, T_1 , T_2 and T are the initial, final and current temperature, t is the time and τ is the time constant. At $t = \tau$, the change in the temperature is 63.2% of the net change in temperature.

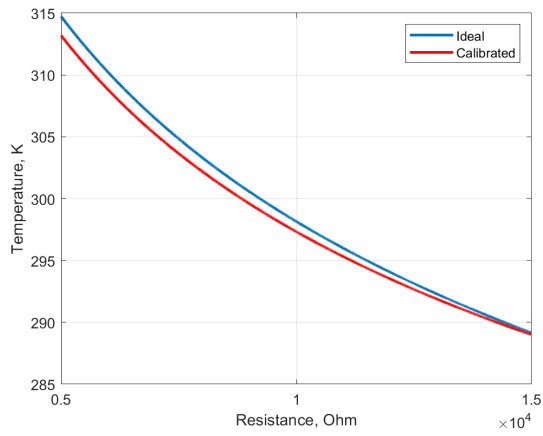
$$T - T_1 = (T_2 - T_1)(1 - e^{-1}) \implies \frac{T - T_1}{T_2 - T_1} = 1 - \frac{1}{e} = 0.632$$

The thermistors were subjected to a sudden drop in temperatures and their response curves were plotted using the Steinhart-Hart coefficients. The time constant for each of the three thermistors was determined from their response to the temperature drop. These were then analysed to determine if they were suitable to be used in a UAV-based atmospheric study.

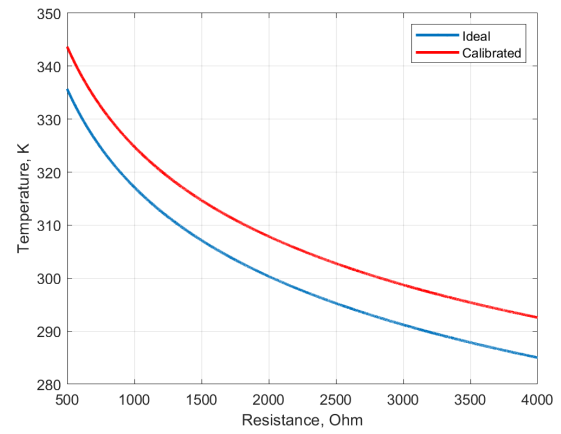
Results

8.1 Calibration

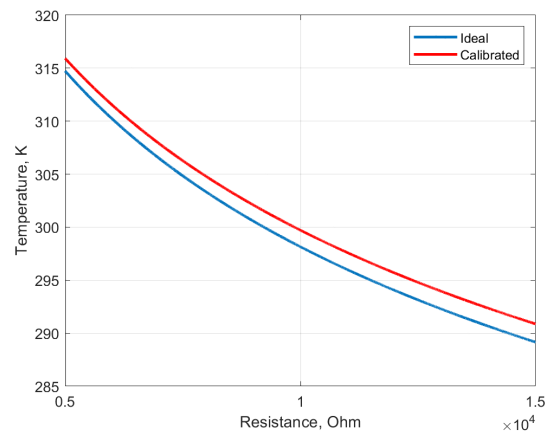
The following are the calibration charts obtained



(a) Thermistor A



(b) Thermistor B



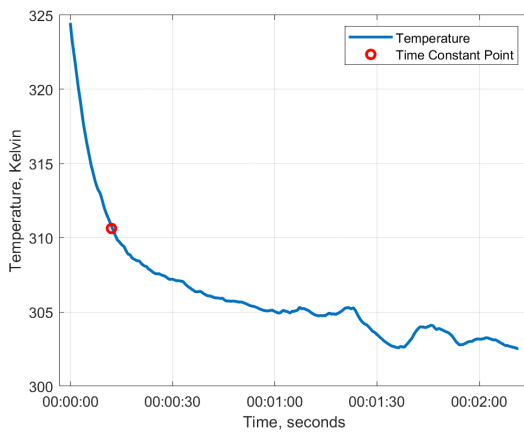
(c) Thermistor C

FIGURE 8.1: Calibration Charts

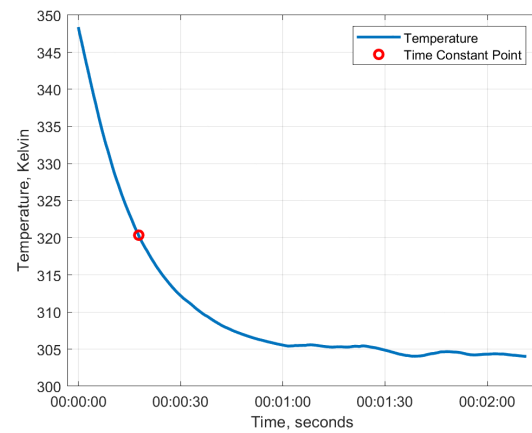
The plots given in Figure 8.1 feature the relationship between the resistance and temperature for each of the three thermistors determined by the calibration, along with the ideal equation for the resistance specified in the datasheet.

8.2 Time Constant Determination

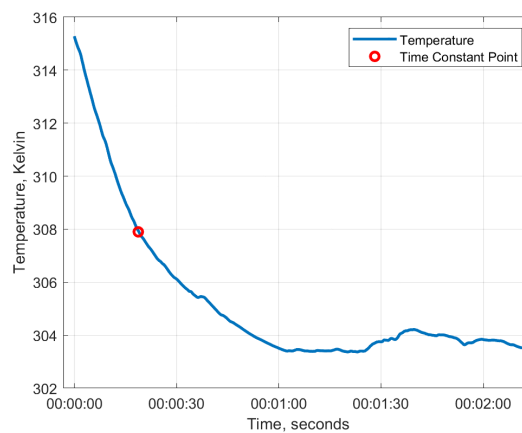
The time constants obtained for each of the three thermistors are given below in figure 8.2 and table 8.1



(a) Thermistor A



(b) Thermistor B



(c) Thermistor C

FIGURE 8.2: Response Curve and Time Constant of Thermistors

Thermistor	Time Constant (s)
A	12
B	17
C	18

TABLE 8.1: Time Constants of Thermistors

The time constants for all the thermistors were above 10 seconds. Thermistor A had the lowest time constant of 12 seconds, it has the smallest sensor tip of all three sensors, owing to its fast response. To use sensors in a UAV-based study, the thermistors tested here are far too slow. At every altitude, we would be able to hover for a short period and to get accurate readings we would require sensors with a significantly smaller time constant.

Conclusions

Three different thermistors were calibrated and then tested to check for their utility in a UAV-based atmospheric study. The temperature sensors are to be used on a drone to capture the temperature at different altitudes as the drone hovers. The time at which the drone can hover at a certain altitude is very small, making the fast response of the sensors used a necessity.

The responsiveness of a thermistor can be characterized in terms of its time constant. For the experiment planned, it was assessed that the time constant should be at least 3 seconds if not less. The thermistors that were tested showed much larger time constants, making them unsuitable for a UAV-based atmospheric study.

9.1 Future Scope

A UAV-based approach to map the temperature profile at higher altitudes in the atmosphere has a lot of scope to collect accurate data, however, it does come with multiple constraints, one of them being the necessity of fast response sensors. The necessity for fast, accurate temperature sensors once fulfilled this experiment can be taken forward to understand the effect of heat transfer processes near the ground at higher altitudes.

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